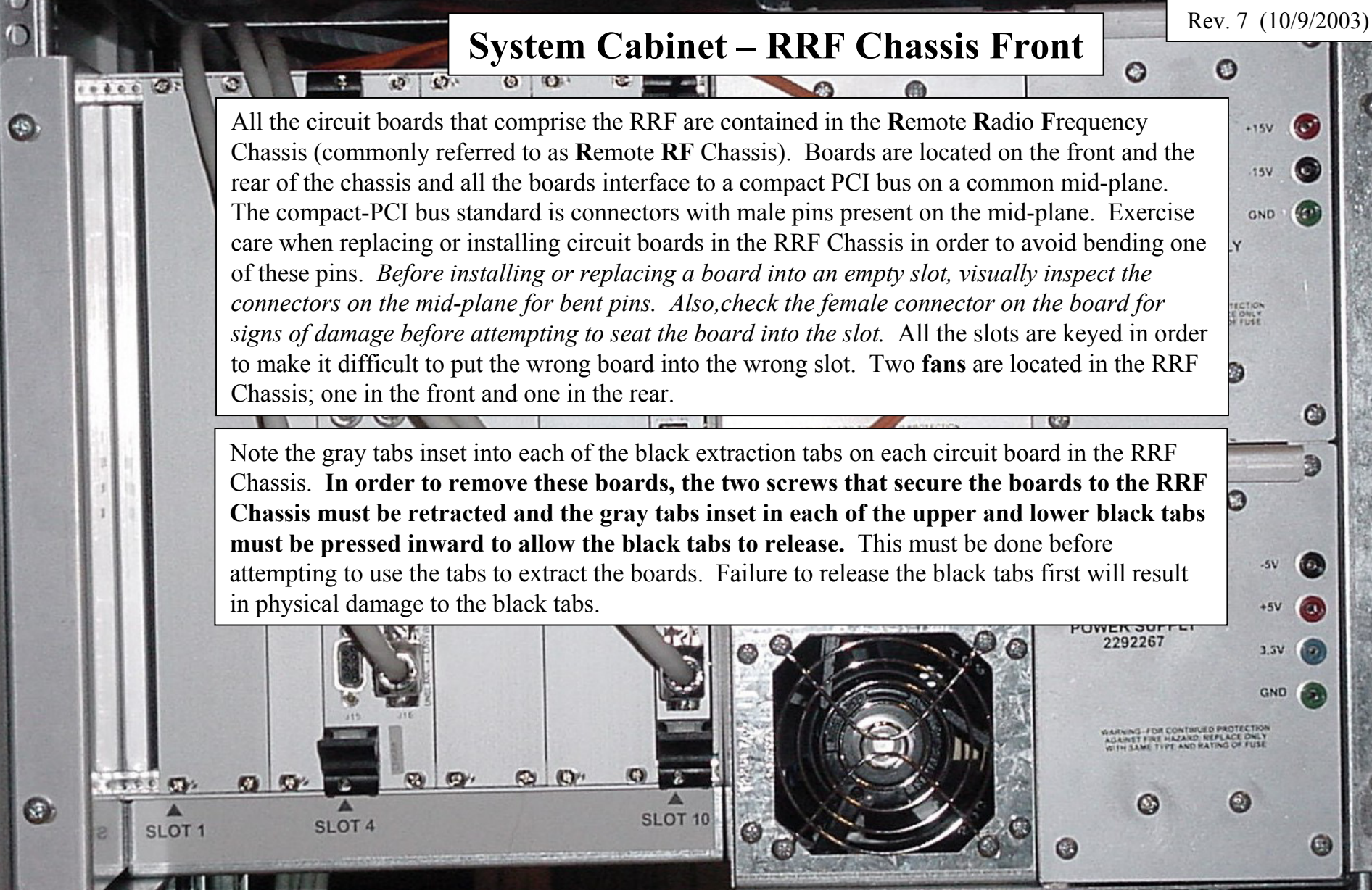


System Cabinet – RRF Chassis Front

All the circuit boards that comprise the RRF are contained in the **Remote Radio Frequency Chassis** (commonly referred to as **Remote RF Chassis**). Boards are located on the front and the rear of the chassis and all the boards interface to a compact PCI bus on a common mid-plane. The compact-PCI bus standard is connectors with male pins present on the mid-plane. Exercise care when replacing or installing circuit boards in the RRF Chassis in order to avoid bending one of these pins. *Before installing or replacing a board into an empty slot, visually inspect the connectors on the mid-plane for bent pins. Also, check the female connector on the board for signs of damage before attempting to seat the board into the slot.* All the slots are keyed in order to make it difficult to put the wrong board into the wrong slot. Two **fans** are located in the RRF Chassis; one in the front and one in the rear.

Note the gray tabs inset into each of the black extraction tabs on each circuit board in the RRF Chassis. **In order to remove these boards, the two screws that secure the boards to the RRF Chassis must be retracted and the gray tabs inset in each of the upper and lower black tabs must be pressed inward to allow the black tabs to release.** This must be done before attempting to use the tabs to extract the boards. Failure to release the black tabs first will result in physical damage to the black tabs.



System Cabinet – RRF Chassis Front

RF-Digital InterFace Board (Slot 10 F)

This board is the primary interface with the MGD Module. Digitized RF transmit, receive, B0, and control data is transferred over the fiber optic cable between the MGD Module to the RRF Module through this board. In the future the RF-DIF will also work with the UTNS to suppress “spike noise” using a digital suppression technique. This function of the RF-DIF Board is not currently enabled. The small coaxial cable connected to J20 provides a special “composite” unblank signal used by the Magnet Monitor. This board is also in the CAN Link chain. CAN Link information from the SCP is transferred through the cable interconnecting to J22 while CAN Link information from the Driver Module is transferred through the cable interconnecting to J23.

Available Diagnostics for Testing the RF-DIF Board

- **RRF Board Level Diagnostics:** DIF BLD
- **RF-DIF Diagnostic:** Master RF-DIF, Slave RF-DIF (if available)
- **RRF Datapaths:** RRF Loopback, RRF Feeder, RRF Checksum

System Cabinet – RRF Chassis Front

Front Panel I/O (FPIO) Board (Slots 4,5 F)

This board provides access to various signals used during troubleshooting. The Master and Slave (if present) Exciter Rho, Theta, and Omega test signals can be accessed from the two BNC ports at the top of the board. The Scope Trigger signal, Master and Slave (if present) Exciter Unblank signals, 10MHz Reference, and DAB OUT signals 5-8 are available from the remaining BNC ports. DAB OUT signals 1-4 are available from sub-D connector J15. Unblank signal (RS-422) is supplied to the Driver Module from J16 on this board. RF signal output from the Exciter Board can be controlled from the **RF Enable** Switch on the front of this board. In normal operation this switch is in the left (ON) position and RF is output from the Exciter Board. Placing this switch in the right (OFF) position prevents RF from being output from the Exciter Board.

+15V
-15V
GND
15V RF
POWER SUPPLY
2292268

WARNING-FOR CONTINUED PROTECTION
AGAINST FIRE HAZARD, REPLACE ONLY
WITH SAME TYPE AND RATING OF FUSE

-5V
+5V
3.3V
GND
5V RF
POWER SUPPLY
2292267

WARNING-FOR CONTINUED PROTECTION
AGAINST FIRE HAZARD, REPLACE ONLY
WITH SAME TYPE AND RATING OF FUSE

SLOT 1

SLOT 4

SLOT 10

System Cabinet – RRF Chassis Front

Note that voltage test points are provided on the front of each of the power supplies.

+5V, 3.3V Digital Power Supply

— Supplies power to all boards utilizing digital logic circuitry.

+15V, -15V RF Power Supply – LED present/missing status indicators for the voltages this supply provides are visible at the rear of the RRF Chassis from the front of the Exciter Board.

-5V, +5V, 3.3V RF Power Supply – LED present/missing status indicators for the voltages this supply provides are visible at the rear of the RRF Chassis from the front of the Exciter Board.

5V DIGITAL
POWER SUPPLY
2292265

15V RF
POWER SUPPLY
2292268

WARNING—FOR CONTINUED PROTECTION
AGAINST FIRE HAZARD, REPLACE ONLY
WITH SAME TYPE AND RATING OF FUSE

5V RF
POWER SUPPLY
2292267

WARNING—FOR CONTINUED PROTECTION
AGAINST FIRE HAZARD, REPLACE ONLY
WITH SAME TYPE AND RATING OF FUSE

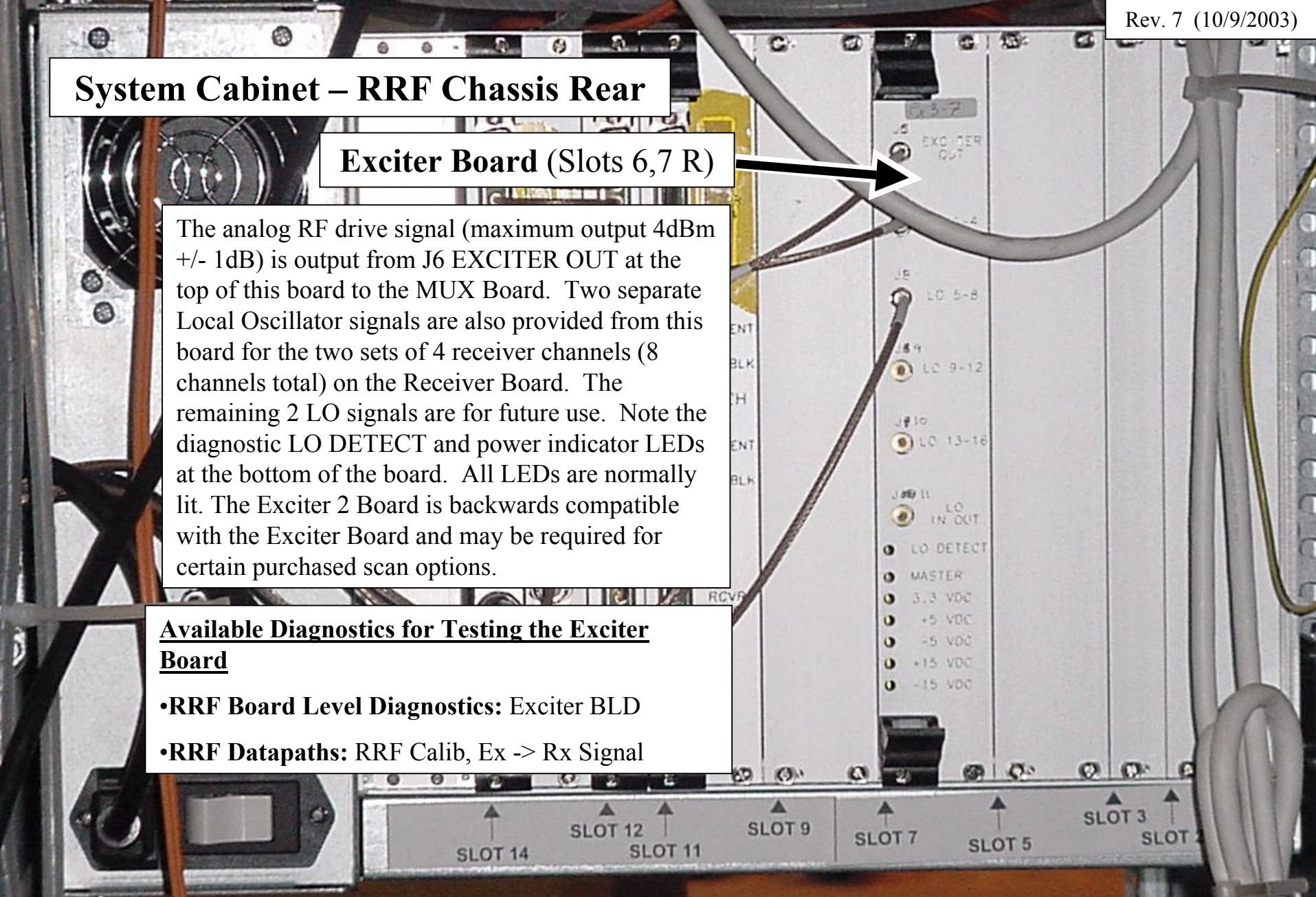
System Cabinet – RRF Chassis Rear

Exciter Board (Slots 6,7 R)

The analog RF drive signal (maximum output 4dBm +/- 1dB) is output from J6 EXCITER OUT at the top of this board to the MUX Board. Two separate Local Oscillator signals are also provided from this board for the two sets of 4 receiver channels (8 channels total) on the Receiver Board. The remaining 2 LO signals are for future use. Note the diagnostic LO DETECT and power indicator LEDs at the bottom of the board. All LEDs are normally lit. The Exciter 2 Board is backwards compatible with the Exciter Board and may be required for certain purchased scan options.

Available Diagnostics for Testing the Exciter Board

- RRF Board Level Diagnostics:** Exciter BLD
- RRF Datapaths:** RRF Calib, Ex -> Rx Signal



System Cabinet – RRF Chassis Rear

Receiver Board (Slots 10,11 R)

Control of the R1 analog receive gain is implemented on this board. This board converts each of the amplified 8 channel receive signals from analog to digital and then forwards the digital data to the RF-DIF Board for transfer to the MGD Module. All LEDs should normally be lit. The Receiver 2 Board is backwards compatible with the Receiver Board.

Available Diagnostics for Testing the Receiver Board

- RRF Board Level Diagnostics: Receiver BLD
- RRF Datapaths: RRF Calib, Ex -> Rx Signal

SLOT 14

SLOT 12

SLOT 11

SLOT 9

SLOT 7

SLOT 5

SLOT 3

SLOT 2

Rev. 7 (10/9/2003)

System Cabinet – RRF Chassis Rear

Universal Transient Noise Suppressor (UTNS) Board (Slot 12 R)

The purpose of this board is to provide receive signal amplification. The UTNS provides about 11dB of gain if the receive R1 value is less than 9 and about 26dB if the R1 value is greater than or equal to 9. Due to the amount of gain provided, bypassing the UTNS out of the circuit during customer scanning, while possible, is usually not practical. Spike noise detection on systems equipped with UTNS or UTNS II is accomplished using the external LX TNS assembly (2 units if system has 8-channel multicoil option) mounted in the bottom of the System Cabinet. The UTNS and UTNS II do not perform spike noise detection and suppression. The UTNS 3 does perform spike noise suppression. This unit is discussed on the next slide. The J30 SMB port on the UTNS is provided to allow monitoring of the Head/Body/MC1 analog receive signal, using an oscilloscope, for spike noise peaks.

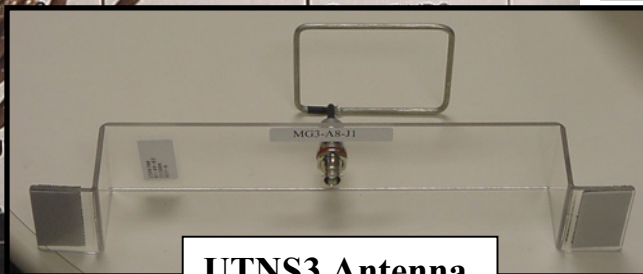
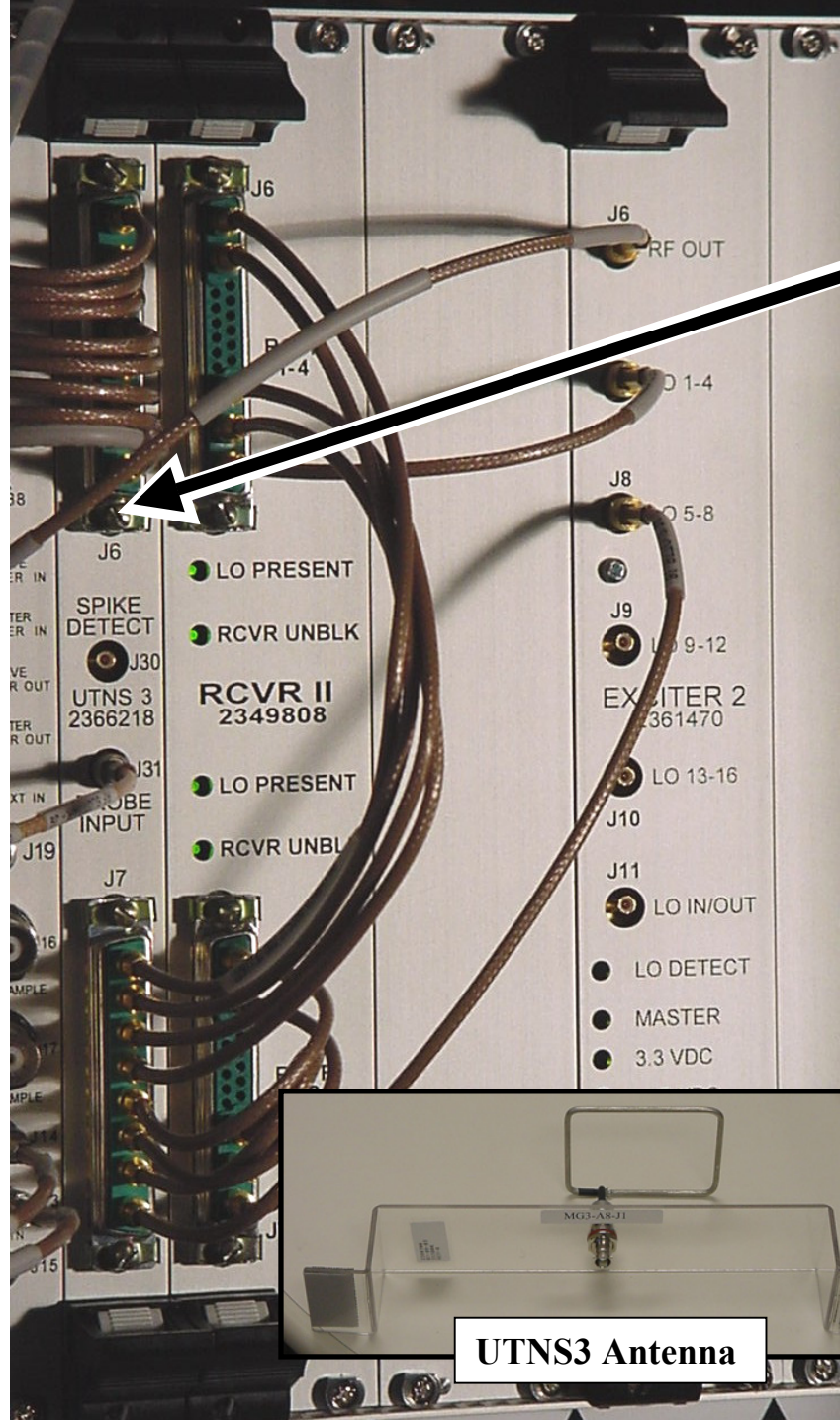
Available Diagnostics for Testing the UTNS Board

- RRF Board Level Diagnostics: UTNS BLD
- RRF Datapath Diagnostics: UTNS/Rx Gain

System Cabinet – RRF Chassis Rear

Universal Transient Noise Suppressor 3 (UTNS3) Board (Slot 12 R)

The UTNS3 is functionally equivalent to the UTNS with the exception that spike noise suppression now is performed on this board. Spike noise is detected from an antenna mounted inside the Rear Pedestal. The signal is then transferred to the UTNS3 through the J31 port marked, "PROBE INPUT". Spike noise is then "notched out" of each of the receive channels in use. This detection method permits the UTNS3 to detect spikes over a broad band of frequencies and thus it is capable of being utilized over a wide range of field strengths. Software on the **CSD** under Utilities can be accessed to view the cumulative number of spike noise counts, reset the cumulative counter, and to enable and disable spike noise suppression on individual channels. A standalone script is also available for accomplishing these functions. See the previous slide concerning the UTNS for more operational details concerning the UTNS3. The UTNS3 replaces the old-style TNS.



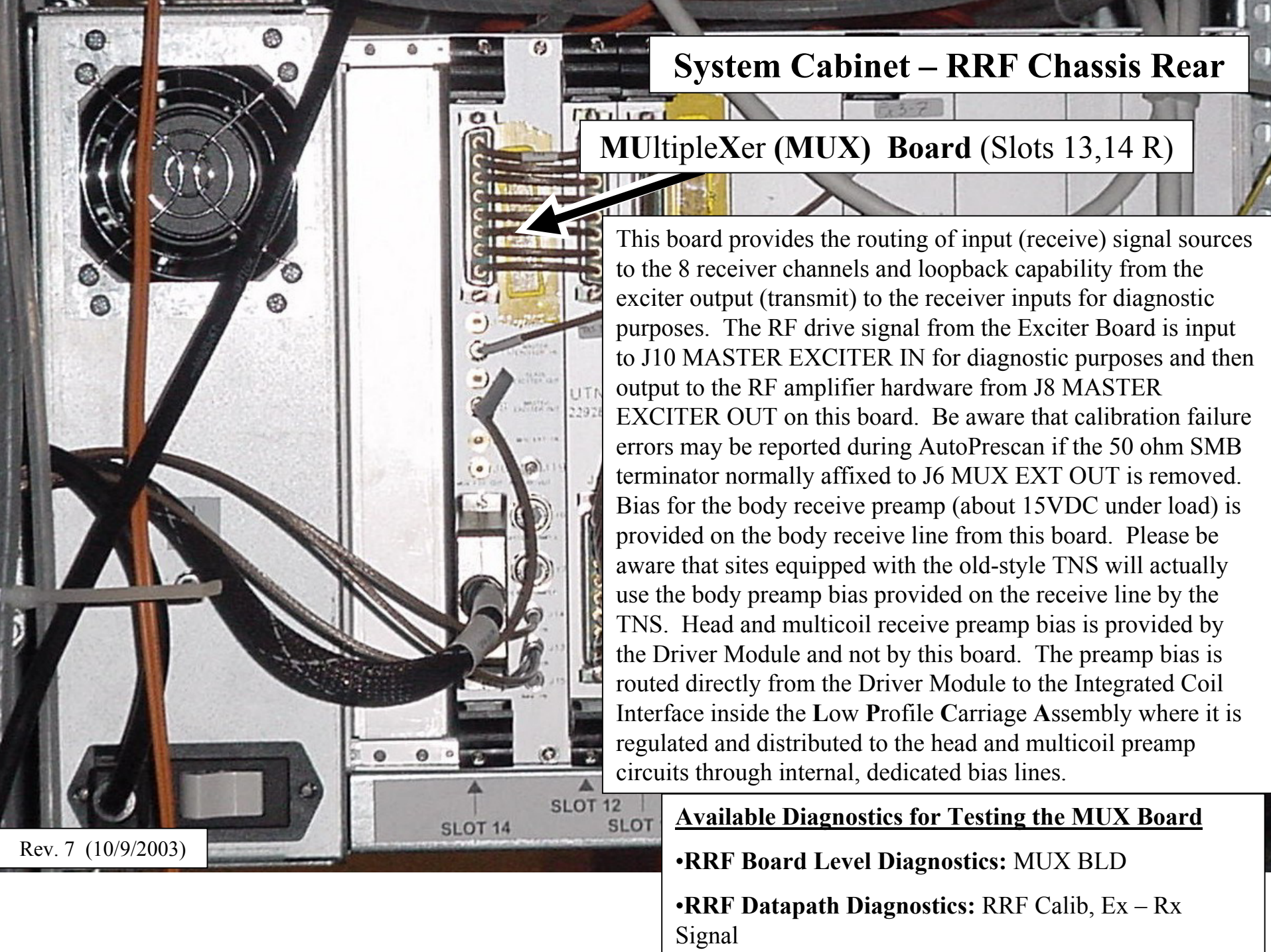
UTNS3 Antenna

Available Diagnostics for Testing the UTNS Board

- RRF Board Level Diagnostics: UTNS BLD
- RRF Datapath Diagnostics: UTNS/Rx Gain

System Cabinet – RRF Chassis Rear

MultipleXer (MUX) Board (Slots 13,14 R)



This board provides the routing of input (receive) signal sources to the 8 receiver channels and loopback capability from the exciter output (transmit) to the receiver inputs for diagnostic purposes. The RF drive signal from the Exciter Board is input to J10 MASTER EXCITER IN for diagnostic purposes and then output to the RF amplifier hardware from J8 MASTER EXCITER OUT on this board. Be aware that calibration failure errors may be reported during AutoPrescan if the 50 ohm SMB terminator normally affixed to J6 MUX EXT OUT is removed. Bias for the body receive preamp (about 15VDC under load) is provided on the body receive line from this board. Please be aware that sites equipped with the old-style TNS will actually use the body preamp bias provided on the receive line by the TNS. Head and multicoil receive preamp bias is provided by the Driver Module and not by this board. The preamp bias is routed directly from the Driver Module to the Integrated Coil Interface inside the Low Profile Carriage Assembly where it is regulated and distributed to the head and multicoil preamp circuits through internal, dedicated bias lines.

Available Diagnostics for Testing the MUX Board

- RRF Board Level Diagnostics: MUX BLD
- RRF Datapath Diagnostics: RRF Calib, Ex – Rx Signal